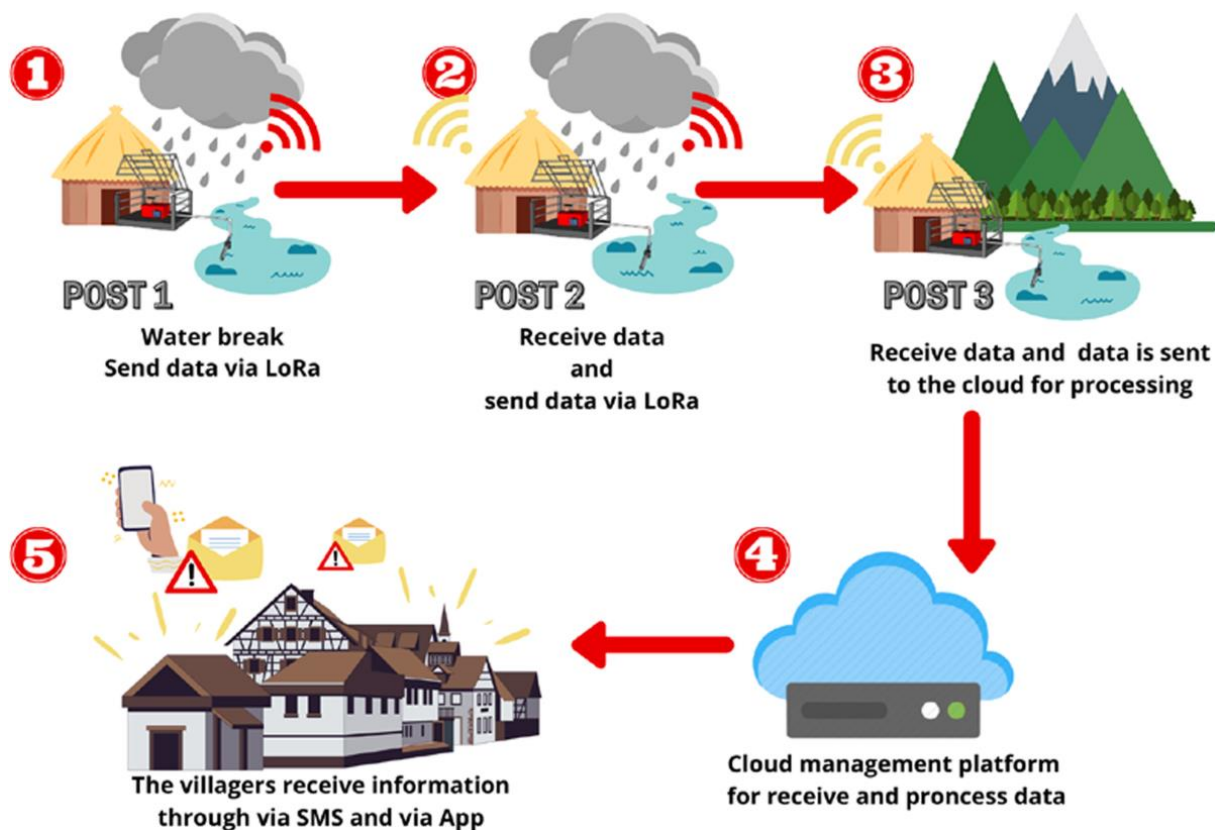
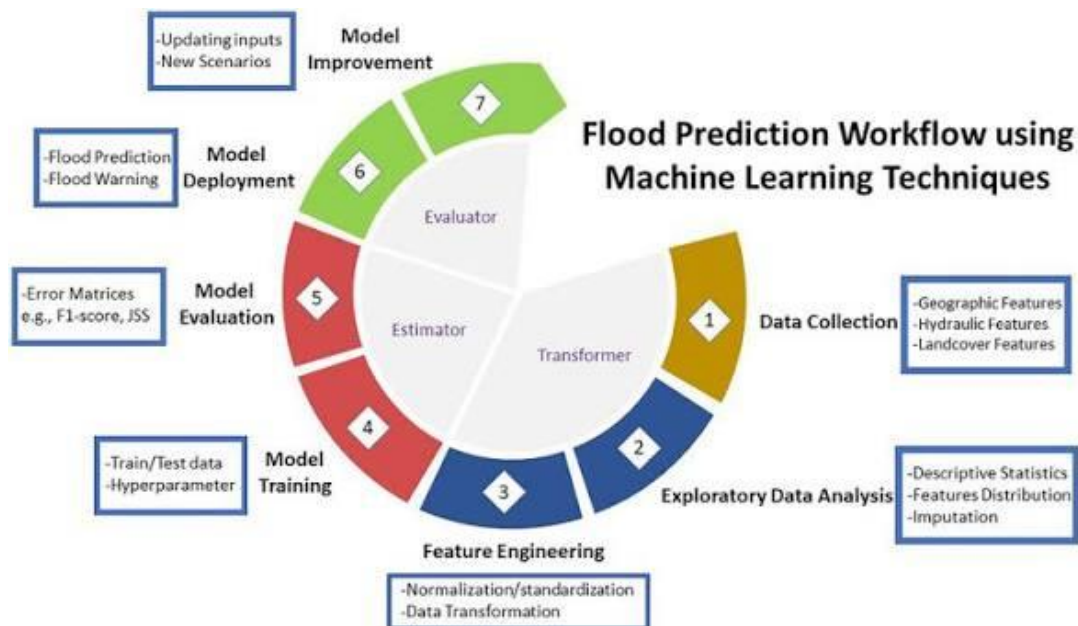


Solution Architecture

Date	7 July, 2026
Team ID	
Project Name	Machine Learning-Based Flood Prediction and Early Warning System
Maximum Marks	5 marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description/Role	Technologies Used
Presentation Layer	Collects applicant details and displays prediction.	Flask, HTML, CSS
API Gateway	Receives prediction requests from the web application and forwards them to the prediction engine.	Flask
Input Processing	Validates user input and preprocesses weather data before prediction.	Python, Scikit-learn, XGBoost
Core Logic Service	Uses trained Machine Learning models to predict flood occurrence based on weather parameters.	Python, Scikit-learn, XGBoost
Dataset Storage	Stores historical weather and flood datasets used for training and testing..	CSV, Pickle
Prediction Result Module	Displays the flood prediction result (Flood Likely / No Flood) with confidence level.	Flask, HTML, CSS